

Michael Serrano

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Education

Massachusetts Institute of Technology, Cambridge, MA	GPA: 4.6/5.0
Master of Engineering Computer Science, Economics, and Data Science	January 2027
Bachelor of Science Physics, Computer Science, Economics, and Data Science	June 2026

Technical Skills

Languages: Python, Go, TypeScript, Rust, R, SQL, C
System Design: Dual-database design, async queues, data pipelines, self-healing workers, rate limiting, reverse proxies, horizontal scaling, vector search, SSE
Tools & Infra: Postgres, Redis, Docker, GitHub Actions, Railway, Cloudflare, Sentry, Playwright, AWS, Celery, Airflow, Nginx, MinIO
Protocols & Auth: REST APIs, MCP, OAuth 2.0 PKCE, JWT, AES-256-GCM encryption, OIDC, RBAC
ML/AI: PyTorch, TensorFlow, scikit-learn, LangChain, Claude API, Snorkel, pgvector, NumPy, Pandas, PySpark

Work and Research Experience

Co-Founder, Software Engineer | Rindler (YC S26) (*San Francisco, CA*) November 2025 — Present

- Built Rindler, a translation layer between AI agents and the live web, turning human websites into robust MCP tools for navigation, transactions, and multi-step tasks
- Built a browser execution platform with priority queues, ephemeral workers, and self-healing retries
- Split control and data planes, added observability and rate limits, and hardened the platform for production use
- Rindler achieved **4x faster** task completion (407s→98s) and **6x cheaper** cost (\$1.54→\$0.26) than OSS agents

LLM Researcher | MIT CSAIL (*Cambridge, MA*) January 2025 — Present

- Built and deployed ConceptQ for MIT course staff, turning materials into editable concept graphs and Bloom-aligned quiz questions with RAG grounding
- Productionized a multi-service research platform and release pipeline so course staff could ship changes reliably
- Built real-time quiz generation and secure instructor workflows for classroom use
- Designed the storage and monitoring layer with deduplication to support retrieval, debugging, and scale

Machine Learning Intern | Roblox (*San Mateo, CA*) May — August 2025

- Developed an LSH-minhash + logistic regression pipeline for alt-account detection across **150B user-pairs**
- Introduced Snorkel-based weak supervision, cutting the model release cycle from **months to 1 hour**
- Improved alt-account recall from **94% to 96%** at 90% precision via refined candidate generation
- Rebuilt and documented a defunct training pipeline, restoring rapid retraining and PR-curve evaluation

Machine Learning Intern | Better Off Social (*New York, NY*) June — August 2024

- Trained a GCN to partition event attendees into compatible groups; **99.2%** satisfaction rating in production
- Built a compatibility model from demographic and personality features for downstream matching
- Refactored matching algorithm codebase and documentation, yielding a **54x speed-up**

Algorithmic Game Theory Researcher | MIT Sloan (*Cambridge, MA*) January — May 2023

- Studied reinforcement-learning pricing agents and how prior market knowledge drives emergent collusion
- Analyzed how prior market knowledge shifts Nash equilibrium outcomes between firms

Leadership and Extracurriculars

Teaching Assistant | Modeling with ML & Intro to ML Courses (*Cambridge, MA*) February 2024 — May 2025

- Taught fundamentals of modeling, supervised learning, and unsupervised learning
- Helped adapt curriculum and assessments to better test understanding despite widespread GenAI use

Vice President and Treasurer | Sloan Business Club (*Cambridge, MA*) January 2024 — December 2025

- Managed club expenditures and a \$20,000 budget for events, marketing, and recruitment
- Led external speaker outreach and event logistics

VP of Finance, Risk Management Director | Beta Theta Pi (*Cambridge, MA*) January 2023 — December 2025

- Managed a **\$400,000** budget for house upkeep, programming, and asset management
- Launched a flex-spending system for house improvements, increasing member retention